

Problem Set 2 – Solutions

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Question 1

Let $\mathcal{F}_t$ denote the filtration (sigma algebra) generated by $y_t, y_{t-1}, y_{t-2}, \dots$. Note that by repeated substitution,

$$\begin{aligned}\sigma_t^2 &= \nu + \alpha u_{t-1}^2 + \beta \sigma_{t-1}^2 \\ &= \nu + \alpha u_{t-1}^2 + \beta(\nu + \alpha u_{t-2}^2 + \beta \sigma_{t-2}^2) \\ &= \sum_{\ell=1}^{\infty} \beta^{\ell-1} \{\nu + \alpha u_{t-\ell}^2\} \\ &= \sum_{\ell=1}^{\infty} \beta^{\ell-1} \{\nu + \alpha(y_{t-\ell} - \rho y_{t-\ell-1})^2\} \\ &\in \mathcal{F}_{t-1}.\end{aligned}$$

Hence, u_t is a martingale difference sequence with respect to $\mathcal{F}_t$:

$$u_t = y_t - \rho y_{t-1} \in \mathcal{F}_t,$$

$$E(u_t \mid \mathcal{F}_{t-1}) = E(\sigma_t \varepsilon_t \mid \mathcal{F}_{t-1}) = \sigma_t \underbrace{E(\varepsilon_t \mid \mathcal{F}_{t-1})}_{=E(\varepsilon_t)=0} = 0.$$

Part (i)

By stationarity,

$$\begin{aligned}E(\sigma_t^2) &= \nu + \alpha E(u_t^2) + \beta E(\sigma_t^2) = \nu + \alpha E(\sigma_t^2) \underbrace{E(\varepsilon_t^2)}_{=1} + \beta E(\sigma_t^2) \\ \implies E(\sigma_t^2) &= \frac{\nu}{1 - \alpha - \beta}.\end{aligned}$$

Also, since $\{y_t\}$ is an AR(1) process with martingale difference (and thus white noise) errors, the usual formula applies:

$$\text{Var}(y_t) = \frac{E(u_t^2)}{1 - \rho^2} = \frac{E(\sigma_t^2)}{1 - \rho^2}.$$

Part (ii)

Since u_t is a martingale difference sequence with respect to $\mathcal{F}_{t-1}$,

$$E(y_t \mid \mathcal{F}_{t-1}) = E(\rho y_{t-1} + u_t \mid \mathcal{F}_{t-1}) = \rho y_{t-1} + E(u_t \mid \mathcal{F}_{t-1}) = \rho y_{t-1}.$$

Moreover, because $\sigma_t \in \mathcal{F}_{t-1}$ and ε_t is independent of $\mathcal{F}_{t-1}$,

$$\text{Var}(y_t \mid \mathcal{F}_{t-1}) = \text{Var}(u_t \mid \mathcal{F}_{t-1}) = \text{Var}(\sigma_t \varepsilon_t \mid \mathcal{F}_{t-1}) = \sigma_t^2 \underbrace{\text{Var}(\varepsilon_t \mid \mathcal{F}_{t-1})}_{=\text{Var}(\varepsilon_t)=1} = \sigma_t^2.$$

Part (iii)

By part (ii) and Gaussianity of ε_t , we have

$$(y_t \mid \mathcal{F}_{t-1}) \sim N(\rho y_{t-1}, \sigma_t^2).$$

Thus, the log likelihood function equals

$$\mathcal{L}(\rho, \nu, \alpha, \beta) = -\frac{T}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^T \left\{ \log(\bar{\sigma}_t^2) + \frac{(y_t - \rho y_{t-1})^2}{\bar{\sigma}_t^2} \right\},$$

where $\bar{\sigma}_t^2$ is given by the recursion

$$\bar{\sigma}_0^2 = \frac{\nu}{1 - \alpha - \beta}, \quad \bar{\sigma}_t^2 = \nu + \alpha(y_{t-1} - \rho y_{t-2})^2 + \beta \bar{\sigma}_{t-1}^2, \quad t = 1, \dots, T.$$

Question 2

See the Matlab script `ps2_q2_solution.m` on the course website, along with the associated Matlab functions. (Note: Unlike me, you are expected to explain your code, report relevant numbers/figures, and discuss the results in your assignment.)

Question 3

Part (i)

Since the ACF of an MA(1) process is zero for lags $\ell > 1$, the spectral density of $\{y_t\}$ equals

$$s_y(\omega) = \frac{1}{2\pi} \left\{ \Gamma_y(0) + 2 \sum_{\ell=1}^{\infty} \cos(\omega\ell) \Gamma_y(\ell) \right\} = \frac{1}{2\pi} \{ \Gamma_y(0) + 2(\cos \omega) \Gamma_y(1) \}, \quad \omega \in [-\pi, \pi].$$

Since the spectral density is nonnegative everywhere, we have

$$\Gamma_y(0) + 2(\cos \omega) \Gamma_y(1) \geq 0 \quad \text{for all } \omega \in [-\pi, \pi].$$

The result in the problem text follows from $\cos \omega \in [-1, 1]$, with $\cos(0) = -\cos(\pi) = 1$.

Part (ii)

Consider now a purely non-deterministic, covariance stationary process $\{y_t\}$ with autocovariances $\Gamma_y(\ell) = 0$ for $\ell \geq 2$. By the Wold decomposition theorem, this process has an MA(1) representation. However, we just proved that it must then be the case that $|\Gamma_y(1)| \leq \Gamma_y(0)/2$. The reason is simple: This inequality is necessary and sufficient for the autocovariance function to be positive semidefinite (see Brockwell & Davis, 1991, 2nd ed., Example 1.5.1, p. 28).

Question 4

Part (i)

Define $z_t \equiv y_t - y_t^\dagger$ and the $(n_y + n_x)$ -dimensional process $u_t \equiv (z_t', x_t')'$. By definition of the best linear predictor $y_t^\dagger$, we have $\text{Cov}(z_t, x_\tau) = 0_{n_y \times n_x}$ for all t, τ . Hence $s_{zx}(\omega) = 0$ for all $\omega \in [-\pi, \pi]$, so

$$s_u(\omega) = \begin{pmatrix} s_z(\omega) & 0_{n_y \times n_x} \\ 0_{n_x \times n_y} & s_x(\omega) \end{pmatrix}.$$

Now define the $(n_y + n_x)$ -dimensional process

$$w_t \equiv \begin{pmatrix} y_t \\ x_t \end{pmatrix} = \begin{pmatrix} I_{n_y} & \Theta(L) \\ 0_{n_x \times n_y} & I_{n_x} \end{pmatrix} \begin{pmatrix} z_t \\ x_t \end{pmatrix} = \Psi(L) u_t, \quad \Psi(L) \equiv \begin{pmatrix} I_{n_y} & \Theta(L) \\ 0_{n_x \times n_y} & I_{n_x} \end{pmatrix}.$$

By the linear filter formula for spectral density matrix functions,

$$\begin{aligned}
s_w(\omega) &= \begin{pmatrix} s_y(\omega) & s_{yx}(\omega) \\ s_{xy}(\omega) & s_x(\omega) \end{pmatrix} \\
&= \Psi(e^{-i\omega})s_u(\omega)\Psi(e^{-i\omega})^* \\
&= \begin{pmatrix} s_z(\omega) + \Theta(e^{-i\omega})s_x(\omega)\Theta(e^{-i\omega})^* & \Theta(e^{-i\omega})s_x(\omega) \\ s_x(\omega)\Theta(e^{-i\omega})^* & s_x(\omega) \end{pmatrix}.
\end{aligned}$$

In particular,

$$s_{yx}(\omega) = \Theta(e^{-i\omega})s_x(\omega),$$

so the result in the problem text follows by non-singularity of $s_x(\omega)$.

Part (ii)

By definition, $y_t^\dagger = \Theta(L)x_t$, so the linear filter formula and part (i) give

$$\begin{aligned}
s_{y^\dagger}(\omega) &= \Theta(e^{-i\omega})s_x(\omega)\Theta(e^{-i\omega})^* \\
&= s_{yx}(\omega)s_x(\omega)^{-1}s_x(\omega)(s_{yx}(\omega)s_x(\omega)^{-1})^* \\
&= s_{yx}(\omega)s_x(\omega)^{-1}s_x(\omega)(s_x(\omega)^*)^{-1}s_{yx}(\omega)^* \\
&= s_{yx}(\omega)s_x(\omega)^{-1}s_{xy}(\omega).
\end{aligned}$$

The last equality uses that $s_x(\omega)$ is Hermitian and that $s_{yx}(\omega)^* = s_{xy}(\omega)$.

Part (iii)

$$\begin{aligned}
|K_{yx}(\omega)|^2 &= \left| \frac{s_{yx}(\omega)}{\sqrt{s_y(\omega)s_x(\omega)}} \right|^2 \\
&= \frac{|s_{yx}(\omega)|^2}{s_y(\omega)s_x(\omega)} \\
&= \frac{s_{yx}(\omega)s_{yx}(\omega)^*/s_x(\omega)}{s_y(\omega)} \\
&= \frac{s_{yx}(\omega)s_{xy}(\omega)/s_x(\omega)}{s_y(\omega)} \\
&= \frac{s_{y^\dagger}(\omega)}{s_y(\omega)},
\end{aligned}$$

where the last equality uses part (ii).

Question 5

See the Matlab script `ps2_q5_solution.m` on the course website, along with the associated Matlab functions and data set.

Question 6

See the Matlab script `ps2_q6_solution.m` on the course website, along with the associated Matlab functions and data set.